

NXR-5700/5800

NEXEDGE VHF/UHF Digital & FM Base-Repeater

NXDN™



FOCUSED ON THE FUTURE

Over 7x faster processing speed and 15x more memory capacity than the previous models, these new NEXEDGE repeaters represent a breakthrough in performance. Extensive data storage means they can support everything from analog/digital conventional systems up to a highly sophisticated NEXEDGE Generation2 (Gen2) multi-site digital trunked network. Stay ahead of the curve, with cutting-edge communications.



Features

- Wideband Coverage
- 25/5/0.5 W RF Output Power (100% Duty Cycle)
- Two-Digit Numeric Display
- LED Status Indicators
- USB 2.0 Type-B Interface
- IP LAN/WAN Connectivity
- Ethernet Network Interface
- 6 Programmable Function Keys
- 0.3 W Front Panel Speaker
- 3 W External Speaker Audio
- Audio Volume Control
- Program / Modem Interface
- Remote Termination Interface
- Programmable AUX I/O's
- DTMF Remote Control
- Flash Firmware Upgrading
- Remote System Firmware Updates
- Telephone Interconnect Option

Digital – General

NXDN Digital Air Interface	UID & GID Validation
AMBE+2TM VOCODER	NXR Over-the-Air Alias
6.25 & 12.5 kHz Bandwidth	SNMP Protocol Ready
Built-In 0.5 ppm TCXO	FER (Frame Error Rate) / RSSI Output

Digital – Trunking Mode

NEXEDGE Gen2 Network (NXDN Type-C Trunking (Gen1))	Control Channel Rotation
Transmission Trunked Mode	Cross-Busy
Message Trunked Mode	Failsoft Mode
Busy Call Queuing	NXDN Traffic Channel Sharing
Call Queue Pre-emption	ESN Validation
Late Entry (UID & GID)	Auto-Roaming / Registration
Control / Traffic Channel Switching	Wide Area All Group Call

Digital – Conventional Mode

Mixed FM / Digital Operation	Digital Voting
Conventional IP Networks	RF Link
Site Roaming Capability	Digital Simulcast

FM Analog Mode

16 QT/DQTs Repeater Control Built-in	EIA Voter Tone Generation
Hang Timer / Time Out Timer / CW ID	External LTR® Controller Interface
External FM Controller Interface	External MPT1327 Controller Interface

Accessories

All accessories may not be available in all markets.
Contact an authorized KENWOOD dealer for details and complete list of all accessories.

KMC-65M
Standard Microphone



KMC-66M
DTMF Microphone



KXK-3M3
High Stability
Internal Reference
Oscillator



Specifications

General	NXR-5700	NXR-5800
Frequency Range		
Type 1	136 - 174 MHz	450-520 MHz
Type 2		400-470 MHz
Type 3		350-400 MHz
Channel Spacing		
Analog	25*/12.5 kHz	
Digital	12.5/6.25 kHz	
PLL Channel Step	2.5 / 3.125 / 5 / 6.25 kHz	3.125 / 5 / 6.25 kHz
Power Supply	13.6 V DC (10.8 - 15.6 V DC)	
Operating Temperature	-22 °F to + 140 °F (-30 °C to +60 °C)	
Frequency Stability	± 0.5 ppm	
Antenna Impedance	50 Ω	
Dimensions	(W x H x D) Projections Not Included 19.02 x 1.73 x 13.03 in (483 x 44 x 331 mm)	
Weight Radio	11 lb (5 kg)	
FCC ID		
Type 1	K44474500	K44474600
Type 2		K44474601
IC Certification		
Type 1	282F-474500	
Type 2		282F-474601

*Analog 25 kHz is not included in the models sold in the USA or US territories.
Measurements made per CAI measurement procedures (digital) and TIA-603 (analog); are measured according to applicable standards. Specifications shown are typical and subject to change without notice, due to advancements in technology.
Details and timing of firmware and software updates are subject to change without notice.

Receiver	NXR-5700	NXR-5800
Sensitivity		
Digital @ 6.25 kHz (3% BER)		0.27 µV
Digital @ 12.5 kHz (3% BER)		0.33 µV
Analog (12 dB SINAD)		0.30 µV
Selectivity		
Analog @ 25* kHz	92 dB	86 dB
Analog @ 12.5 kHz	84 dB	80 dB
FM Hum & Noise		
Analog @ 12.5kHz		50 dB
Analog @ 25kHz		55 dB
Intermodulation		85 dB
Spurious Rejection		100 dB
Audio Distortion	Less than 2% (at 0.3 W)	
Audio Output	3 W (at 4 Ω Less than 5 % distortion)	

Transmitter	NXR-5700	NXR-5800
RF Power Output High / Low	25 / 5 / 0.5 W	
Max Duty Cycle	100%	
Spurious & Harmonics	73 dB	
FM Hum & Noise		
Analog @ 12.5kHz		50 dB
Analog @ 25kHz		55 dB
Audio Distortion	Less than 1% at 1000 Hz	
Emission Designator	16K0F3E, 11K0F3E, 8K30F1E, 8K30F1D, 8K30F7W, 4K00F1E, 4K00F1D, 4K00F7W, 4K00F2D	

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MIL-STD & IP

MIL Standard	MIL 810C Methods/Procedures	MIL 810D Methods/Procedures	MIL 810E Methods/Procedures	MIL 810F Methods/Procedures	MIL 810G Methods/Procedures
High Temperature	501.1/Procedure I, II	501.2/Procedure I, II	501.3/Procedure I, II	501.4/Procedure I, II	501.5/Procedure I, II
Low Temperature	502.1/Procedure I	502.2/Procedure I	502.3/Procedure I	502.4/Procedure II	502.5/Procedure II
Temperature Shock	503.1/Procedure I, II	503.2/Procedure I, II	503.3/Procedure I, II	503.4/Procedure I, II	503.5/Procedure I

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